

Showing Your Work in 2026

A Structural Specification for Reproducible Scientific Computation

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1. What “Showing Your Work” Currently Means

A physicist completes a computation. The computation derives a physical quantity — say, the primordial helium abundance — from a chain of input values and intermediate calculations. The physicist writes a paper. The paper is typeset in LaTeX. The equations are formatted. The derivation is described in prose.

The prose reads something like this: “Using the Planck 2018 value of the baryon density parameter $\Omega_b = 0.0490 \pm 0.0006$, we derive the baryon-to-photon ratio η via the standard relation involving the CMB photon number density (Fixsen 2009) and the critical density. Substituting into the Pitrou et al. (2018) BBN fitting formulae, we obtain $Y_p = 0.2486$, in reasonable agreement with the Aver et al. (2015) measurement of 0.2449 ± 0.0040 .”

The reader who wants to verify this result must perform the following steps. Find the Planck 2018 paper. Locate the Ω_b value in its tables. Find the Fixsen 2009 paper. Locate the CMB temperature value. Calculate the photon number density from the temperature using the standard formula, which requires knowing that formula and looking up $\zeta(3)$. Find the critical density, which requires the Hubble constant and the gravitational constant, each of which requires its own lookup. Compute η from these inputs. Find the Pitrou 2018 paper. Locate the fitting coefficients. Apply the fitting formula. Check whether the result matches the paper’s stated value.

This process has specific failure modes. The reader may copy a value incorrectly from one of the cited papers — transcription error. The paper may use a symbol that means

different things in different conventions and not state which convention is intended — notational ambiguity. The paper may skip steps the author considered obvious — derivation gaps. The paper may have used a specific numerical method with specific parameters and not reported them — unstated numerical choices. The computation may have been performed in double-precision floating point, where rounding errors accumulated across the chain and the final digits depend on the implementation rather than on the physics — floating-point artifact.

These are not descriptions of bad practice. This is standard practice. The best papers in the best journals follow this format. The format was designed for an era when computation meant pen-and-paper derivation and publication meant printed pages. In that era, prose description of a computation was the best available technology for communicating the computation to others. The format served that era.

The era is over. Computation is performed by machines. Storage is effectively unlimited. Distribution is instantaneous and global. The infrastructure that made prose description the only viable format no longer constrains us. What constrains us is convention — the inherited practice of describing computations rather than publishing them.

The structural problem is precise. A prose description of a computation is not a computation. It is a set of instructions for performing a computation, written in natural language, subject to ambiguity, and requiring manual re-execution by every reader who wishes to verify the result. The distance between the description and the computation is the space where irreproducibility lives. Every ambiguity, every unstated choice, every transcription opportunity is a point where two independent readers may arrive at different results from the same paper. The prose description does not prevent this divergence. It enables it.

2. What “Showing Your Work” Should Mean

Showing your work means publishing the work. Not a description of the work. The work itself — the inputs, the derivations, the outputs, the comparisons, the failures — in a form that anyone can run, check, and extend without manual re-derivation.

The standard has five layers. Each layer is necessary. Each has a specific structural role. Together they constitute a complete, verifiable, reproducible scientific computation.

Layer 1: The value store. Every input to every computation exists as a named, typed, versioned, machine-readable entry. Each entry carries a unique key that identifies it unambiguously. The value itself, stored as an exact fraction where possible or with stated precision where not. The unit. The number of significant digits. The uncertainty. The source — the specific paper, specific table, specific year from which the value was obtained. Tags for searchability. Notes for methodological context — not buried in a paper’s appendix but attached to the value itself, at the point where anyone encountering the value will see them.

A value store entry for the primordial helium measurement looks like this. Key: `cosmo_yp_measured_v0`. Value: 0.2449. Unit: dimensionless. Digits: 4.

Uncertainty: 0.0040. Source: “Aver, Olive, Skillman 2015. JCAP 07, 011. Primordial helium mass fraction.” Tags: cosmology, BBN, helium, measured. Notes: “ Y_p = primordial ^4He mass fraction. Weighted average of extragalactic HII region measurements.”

Every field serves a purpose. The key enables machine reference — no ambiguity about which value is being used. The exact representation eliminates floating-point accumulation in downstream computations. The source enables tracing — anyone can check where the value came from and verify it against the original publication. The uncertainty enables proper comparison — a prediction that misses by less than the input’s uncertainty is within the measurement’s own limits. The notes capture methodological context at the point of data entry.

The Boltzmann constant illustrates the precision the store supports. Key: `si_boltzmann_constant_v0`. Value: stored as the exact fraction $1380649/10^{29}$. Source: “SI 2019. Exact by definition.” The constant is exact since the 2019 SI redefinition. The store represents it as exact. No floating-point approximation. No loss of precision. The value is what it is, stored as what it is.

A working value store contains thousands of entries spanning every domain the computation touches. The reference implementation described in this paper contains over 4900 value nodes covering quantum electrodynamics, electroweak physics, cosmology, Big Bang nucleosynthesis, CKM mixing, confinement, gravity, and more. Each node in the same format. Each queryable. Each traceable to its source.

Layer 2: The experiment definition. A machine-readable document that declares everything the computation needs and everything it will produce, written before the computation runs.

The definition has four sections. Dependencies: which value store entries are required as inputs, listed by name. Execution plan: which derivations to run and in what order. Expected outputs: every result the derivation should produce, named in advance. Comparisons: every test that will be applied to the results, with the match mode, the expected value, the tolerance, and the reference source — all stated before the computation executes.

This structure embeds a property that is absent from current scientific practice and essential to the standard: pre-registered falsification. Every comparison in the experiment definition is a falsification condition written before the result is known to the runner. The definition states: “I predict this output will match this reference value within this tolerance, and if it does not, the experiment reports FAIL at this specific point.” The prediction is committed. The tolerance is stated. The criterion is mechanical. The outcome — PASS, FAIL, or INFO — is determined by the comparison, not by the author’s interpretation after the fact.

This is Karl Popper’s demarcation criterion — commitment, specificity, priority, acceptance — implemented as data infrastructure. The commitment is the comparison entry. The specificity is the expected value and tolerance. The priority is structural — the definition exists before the run. The acceptance is automatic — the runner prints FAIL when

the criterion is not met, and the author cannot suppress it without altering the definition, which is itself a versioned, auditable document.

Layer 3: The connection bundle. A semantic manifest that declares the experiment's purpose, its scope, and the cross-domain relationships it tests. The bundle lists every input by name and states what the experiment is checking — which physical identities, which cross-domain derivation chains, which hierarchy levels. The connection bundle is the experiment's thesis statement in machine-readable form.

The bundle also serves as the anti-smuggling guard's manifest. Every input the experiment uses must appear in the bundle's input list. An undeclared value cannot enter the derivation because the infrastructure has no mechanism for it to enter. If a value is not in the store and not in the bundle's input list, the runner cannot access it. The structural prevention of hidden parameters is not a policy. It is an architectural constraint.

Layer 4: The run output. Every result produced by the computation is stored as a new value node in the same format as the input values — keyed, typed, sourced to the specific run that produced it. The run output is not a paper. It is not a summary. It is the complete set of values the computation produced, each traceable, each queryable, each available as input to future computations.

This property — output in the same format as input — makes derivation chains continuous. The results of one experiment become the inputs to the next without transcription, format conversion, or manual re-entry. Each run's output is a permanent, versioned record of what the computation produced at that moment with those inputs.

Layer 5: The report. A human-readable summary generated from the run output. Derivation status — how many succeeded, how many errored. Comparison results — predicted value, measured value, miss percentage, pass/fail status for each test. Located failures — which specific test failed, by how much, at which point in the chain. The report is generated from the output, not written separately. It cannot disagree with the output because it is derived from it.

Anatomy of a Value Store Node The Fundamental Data Unit of the Five-Layer Standard

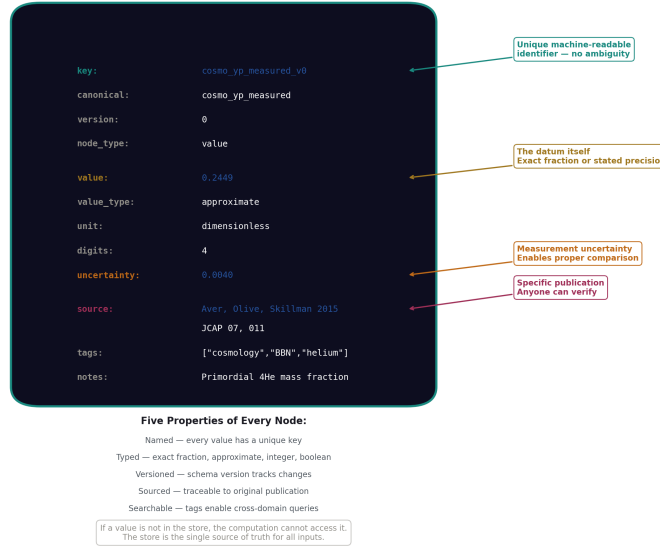


Figure 1: Fig. 8: Value Node Anatomy — A single value store entry dissected, showing every field and its purpose. This is the fundamental data unit of the five-layer standard.

3. A Cross-Domain Example

The five layers become concrete through a specific example. Consider a computation that starts from integer inputs in particle physics and derives nuclear abundances in cosmology — a chain that crosses three domains of physics in a single derivation.

The experiment is called `experiment_bridge_bbn_v0`. Its purpose is to test whether gauge integers from the Standard Model predict primordial nuclear abundances through the cosmological chain. The derivation proceeds as follows. Integer inputs produce the baryon density parameter Ω_b . Ω_b combined with the CMB photon number density produces the baryon-to-photon ratio η . η enters the Big Bang Nucleosynthesis fitting formulae and produces predictions for the primordial helium-4 mass fraction Y_p and the primordial deuterium-to-hydrogen ratio D/H . Separately, the derived cosmological parameters are checked for consistency with the effective number of neutrino species N_{eff} , and the vacuum energy density is derived and compared against observation.

Three domains — particle physics integers, cosmological parameters, nuclear abundances — connected by a single derivation chain. If the integers are wrong, the helium prediction breaks. If the cosmological bridge is wrong, the deuterium prediction breaks. The chain is falsifiable end to end.

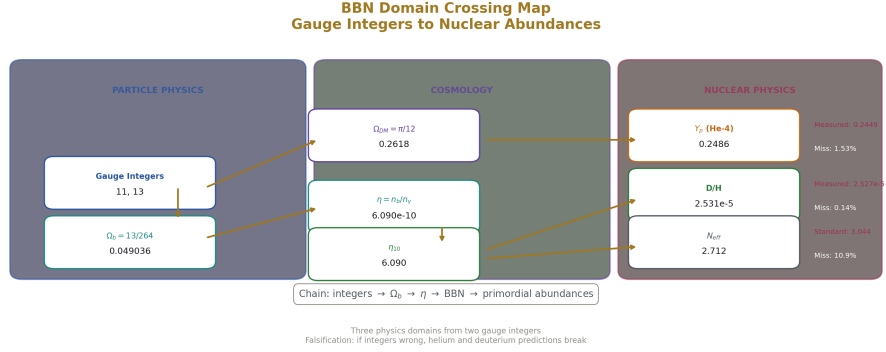


Figure 2: Fig. 1: BBN Domain Crossing Map — Gauge integers flow through three physics domains with actual values at each handoff.

Layer 1 in practice. The experiment draws from twenty-eight named value store entries. Each entry is a specific physical quantity with full metadata. The CMB temperature: `cosmo_t_cmb_v0`, value 2.7255 K, 5 digits, uncertainty 0.0006 K, source “Fixsen 2009. ApJ 707, 916. COBE/FIRAS CMB monopole temperature.” The primordial deuterium measurement: `cosmo_dh_measured_v0`, value 2.527×10^{-5} , 4 digits, uncertainty 0.030×10^{-5} , source “Cooke, Pettini, Steidel 2018. ApJ 855, 102.” The BBN fitting coefficients: `bbn_yp_a_coeff_v0`, value 0.2485, source “Pitrou et al. 2018. Phys.Rept. 754, 1.” Every input named. Every input sourced. Every input machine-readable. A reader verifying the computation does not hunt through citations. They query the value store.

Layer 2 in practice. The experiment definition declares seven derivations in a specific execution order: derive Ω_b from integers, derive Ω_{DE} from flatness, derive η from Ω_b , derive Y_p from η , derive D/H from η , check N_{eff} consistency, derive vacuum energy density. It declares thirteen comparisons — the pre-registered falsification conditions. Each comparison specifies what is being tested, what the output key is, what match mode to use (miss percentage, digit agreement, range check, sigma check), what the expected value is, and where the expected value comes from.

For example, one comparison entry reads: label “D/H digits of agreement,” output key `result_dh_derived_v0`, match mode “digits,” expected “2.53e-5,” digits 3, reference source “Cooke 2018.” This entry states, before the computation runs: the derived deuterium abundance must agree with the Cooke 2018 measurement to three significant

digits. If it does not, the experiment reports FAIL at this point. The falsification condition is pre-registered in the experiment definition. It cannot be adjusted after seeing the result without changing the definition, which is a versioned document.

Another comparison tests the full derivation chain: label “Full chain: integers $\rightarrow$ eta_10 $\rightarrow$ Y_p within 1 sigma,” output key `result_yp_sigma_v0`, match mode “range,” lo 0, hi 2.0. This states: the helium prediction, derived from integers through cosmology through nucleosynthesis, must fall within one standard deviation of the measurement. The sigma value is computed by the derivation and compared against the range by the runner. The result is mechanical.

Layer 3 in practice. The connection bundle for this experiment lists all twenty-eight input values by key and describes the experiment’s scope: “Tests whether gauge integers predict nuclear abundances through the cosmological chain.” The bundle identifies the cross-domain nature of the test explicitly. The inputs span particle physics (gauge integers), cosmology (Planck parameters, CMB temperature, critical density), and nuclear physics (BBN fitting coefficients, measured abundances). The bundle is the manifest — anyone reading it knows what the experiment claims to test, what inputs it uses, and what domains it crosses.

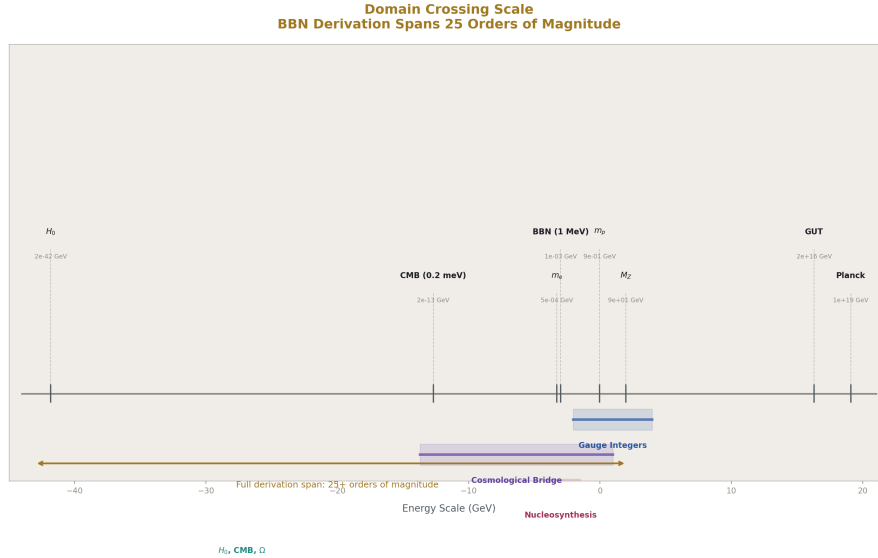


Figure 3: Fig. 2: Domain Crossing Scale — The BBN derivation spans from gauge integers at the Standard Model scale through CMB photons to nuclear abundances, covering 25+ orders of magnitude.

Layer 4 in practice. The run produces fifty-seven output values. Each is stored as a named value node keyed to the specific run. The derived baryon density: `result_omega_b_derived_v0`, value 0.049035637966613, source ex-

periment_bridge_bbn_v0_run011. The derived helium abundance: result_yp_derived_v0, value 0.248643255182356. The derived deuterium abundance: result_dh_derived_v0, value $2.53060482485205 \times 10^{-5}$. The miss percentages, sigma values, intermediate quantities, and consistency checks — all stored, all named, all traceable to the run that produced them.



Figure 4: Fig. 7: Prediction vs Measurement — Integer-derived predictions paired with cosmological measurements for four BBN quantities. Error bars show measurement uncertainty.

Layer 5 in practice. The report prints the complete results. Thirteen comparisons, each with its verdict:

Omega_b from integers versus Planck: predicted 0.049036, measured 0.0490, miss 0.073%. Status: INFO.

D/H digits of agreement: predicted 2.531×10^{-5} , expected 2.53×10^{-5} , three-digit match, miss 0.024%. Status: PASS.

Y_p digits of agreement: predicted 0.2486, expected 0.245, two digits agree but the third does not, miss 1.49%. Status: FAIL.

Full chain integers $\rightarrow \eta \rightarrow Y_p$ within 1 sigma: got 0.936 sigma. Status: PASS.

Full chain integers $\rightarrow \eta \rightarrow D/H$ within 1 sigma: got 0.120 sigma. Status: PASS.

The failure is located precisely. Y_p misses the three-digit agreement target. The miss is

1.49%. The sigma value is 0.94 — within one standard deviation of the measurement, so the full-chain sigma test passes even though the digit test fails. Anyone reading the report knows exactly what succeeded, what failed, and the quantitative gap between prediction and measurement at the failure point.

The summary is generated, not written: seven derivations, zero errors. Four passes, one failure, eight informational. Status: partial. The experiment does not claim success. It reports what it found. The failure is printed with the same mechanical precision as the successes.

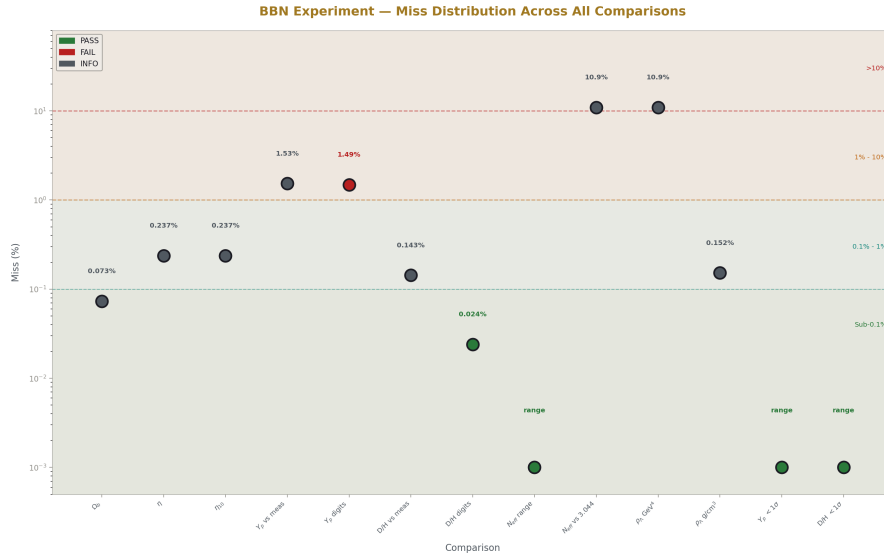


Figure 5: Fig. 6: Miss Distribution — All 13 BBN comparisons plotted on a log scale with threshold regions. Most predictions are sub-percent. The Y_p digit failure is located precisely.

4. The Anti-Smuggling Property

Smuggling is an undeclared input affecting the result of a computation. In current practice, smuggling is easy and usually unintentional. A physicist uses a value they remember from a previous calculation without re-checking it against the current reference. A numerical parameter — integration step count, precision setting, convergence threshold — is chosen and not reported. A fitting parameter is adjusted until the result matches expectation and then presented as derived.

Under the five-layer standard, smuggling is structurally prevented. Every input must exist in the value store. The experiment definition declares every input by name. The connection bundle lists them. The runner loads only declared inputs. If a value is not in the

store, the computation does not silently produce a wrong result. It refuses to run. The infrastructure has no mechanism for an undeclared value to enter the derivation.

The BBN experiment demonstrates this. Twenty-eight inputs are declared in the dependencies section of the experiment definition. The runner loads the value store, checks that each declared dependency exists, and proceeds only if all dependencies are satisfied. If someone added a twenty-ninth input to the derivation code without declaring it in the experiment definition, the runner would access the value store for a key that was never loaded and fail with a traceable error.

Even numerical configuration is stored in the value store. Integration step counts, decimal precision settings, convergence thresholds — these are not buried in source code comments. They are named value nodes with sources and notes. The reference implementation stores `config_euler_step_count_v0` at 10000 with the note: “Increased from 4000 (original) to 10000 for improved two-loop convergence.” The choice is recorded. The change is documented. If changing the step count changes the result, that sensitivity is discoverable because both configurations are stored, both are versioned, and both runs are reproducible.

The anti-smuggling property is not a policy. It is not a guideline that researchers are encouraged to follow. It is an architectural constraint enforced by the infrastructure. The same way a compiler rejects code that references an undeclared variable, the runner rejects computations that reference undeclared values. The mechanism is identical. The consequence is identical. You cannot use what you have not declared.

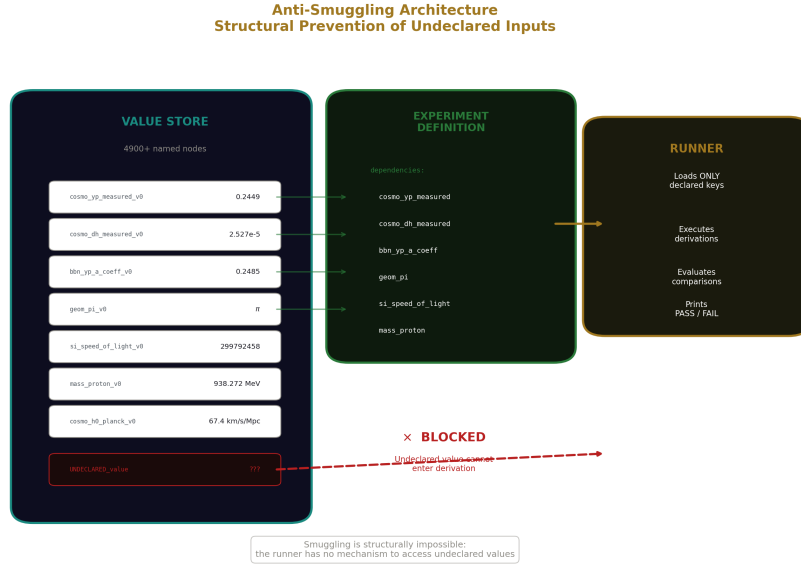


Figure 6: Fig. 4: Anti-Smuggling Architecture — The runner can only access values declared in the experiment definition. Undeclared values are structurally blocked.

5. Pre-Registered Falsification as Infrastructure

The experiment definition’s comparison section deserves specific attention because it implements something that current scientific practice discusses extensively but rarely achieves: pre-registration of falsification conditions.

In current practice, pre-registration is a separate administrative process. A researcher files a pre-registration document with a registry before conducting an experiment. The document states what will be tested and what criteria will determine success or failure. The document exists in a separate system from the experiment itself. Compliance is voluntary and verified after the fact.

Under the five-layer standard, pre-registration is not a separate process. It is the experiment definition. The comparisons section of the definition is the pre-registration. It states what will be tested, what the expected values are, what the tolerances are, and what constitutes pass or fail — and it states all of this as machine-readable data that the runner executes automatically.

The BBN experiment’s definition contains thirteen comparison entries. Each entry was written before the run. Each specifies a mechanical criterion. The runner evaluates each criterion against the computation’s output and prints the verdict. The author does not interpret the results. The runner compares numbers. The verdict is PASS, FAIL, or INFO.

The author can examine the results, investigate the failures, and plan improvements — but the author cannot change the verdict without changing the definition, and the definition is a versioned document whose changes are traceable.

This is stronger than external pre-registration in three specific ways. First, it is structural rather than administrative — the pre-registration is embedded in the computation infrastructure, not filed with a separate registry. Second, it is automatically enforced — the runner evaluates the criteria mechanically, removing the opportunity for post-hoc reinterpretation of what “success” means. Third, it is granular — each comparison is an independent test with its own criterion, so a partial failure is located at a specific point rather than producing an ambiguous overall verdict.

The BBN experiment demonstrates all three properties. It pre-registers thirteen tests. Eleven produce clear verdicts (four PASS, one FAIL, six INFO with quantified miss). The failure is at Y_p digit agreement — located at a specific comparison, with a specific miss of 1.49%. The experiment does not hide this failure. It cannot hide it. The comparison was in the definition. The runner evaluated it. The report printed it.

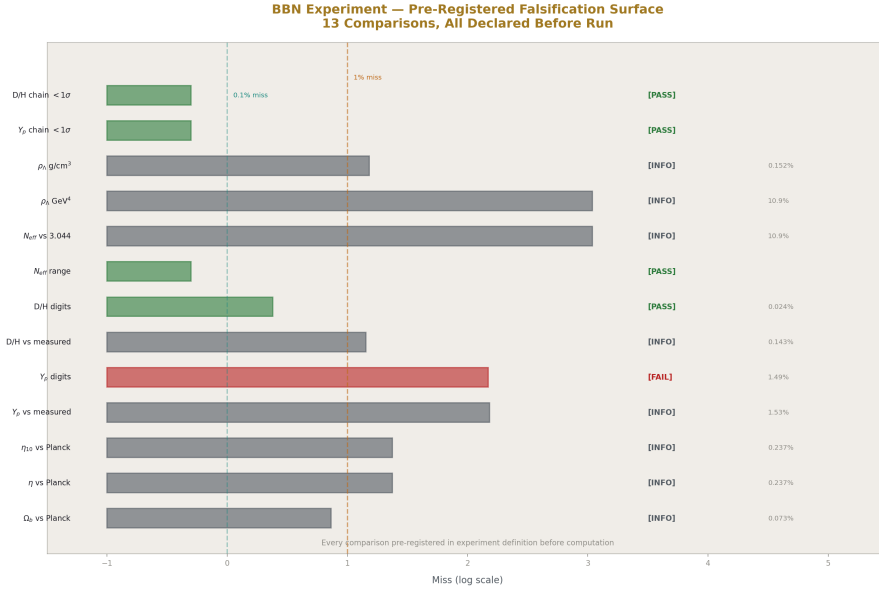


Figure 7: Fig. 3: Comparison Verdict Landscape — All 13 BBN comparisons with their pre-registered pass/fail verdicts and miss percentages on a log scale.

6. The Sensor Measurement Extension

The five-layer standard applies to theoretical computation as described above. It extends to experimental measurement with the instrument replacing the derivation.

Layer 1 is the same — calibration values, instrument parameters, and environmental meta-data stored as named, typed, sourced value nodes. Layer 2 is the measurement protocol — what the instrument will measure, what inputs it requires, what outputs it will produce, and what quality criteria determine whether a reading is valid. Layer 3 is the semantic context — what physical quantity is being measured and how the measurement relates to other measurements. Layer 4 is the raw data — every reading with its timestamp, stored in the same value node format at the instrument’s native sampling rate, with timestamps synchronized to a common clock. Layer 5 is the reduction pipeline — the computation that converts raw readings into published values, itself subject to the five-layer standard.

The critical requirement for sensor measurements: the raw data and the reduction pipeline are published together. The published value is one output of a specific reduction pipeline applied to specific raw readings. Anyone who wants a different output — a different averaging window, different outlier criteria, different weighting — applies their own pipeline to the same raw data. The raw data is the ground truth. The reduction pipeline is a computation. The published value is a result. All three are available.

This is the proposal from CULT-13 made concrete as an engineering specification. CULT-13 argued that CODATA’s averaging process destroys temporal structure, disagreement structure, and contextual correlation. The five-layer standard specifies what replaces the averaging: raw timestamped readings (Layer 4), the reduction methodology as executable code (Layer 5), and the published value as one view of the data rather than the only view. Nothing is destroyed. Everything is preserved. The average can still be computed by anyone who wants it. But the raw structure is available to anyone who wants to ask questions the average cannot answer.

7. What This Standard Costs

The standard requires more work than current practice. The cost is quantifiable.

The value store costs data entry time. Each input value must be entered with its full metadata — source, uncertainty, digits, tags, notes. For a computation with twenty-eight inputs, as in the BBN experiment, this is perhaps one to two hours of careful data entry. The return: the inputs are entered once and reused across every computation that references them. The reference implementation’s 4900-node store was built incrementally. Each new experiment adds a few values and reuses hundreds. The marginal cost of adding a new experiment to an established store is minutes, not hours.

The experiment definition costs planning time. Declaring inputs, outputs, and comparisons in advance requires thinking through the computation before running it. This is not overhead. This is the commitment step. The time spent writing the definition is the

time spent making the prediction specific and testable before the test. Under current practice, this thinking happens informally and incompletely. Under the standard, it happens explicitly and is documented.

The run infrastructure costs implementation time. Building a runner that loads the value store, executes derivations, runs comparisons, and produces reports is an engineering project. The reference implementation exists and is operational — fifty experiments across multiple domains of physics. For a new research group, adopting the standard means either using the existing infrastructure or building a compatible implementation against the specification. This is a one-time cost.

The total cost is small relative to the cost of the science. A single precision measurement experiment at a national laboratory costs millions of dollars and years of effort. The five-layer standard costs days of engineering to implement and hours per experiment to maintain. The cost of not implementing it — irreproducible results, unlocated errors, smuggled parameters, destroyed temporal structure — is paid by every downstream user of the results, repeatedly, for as long as the results are used.

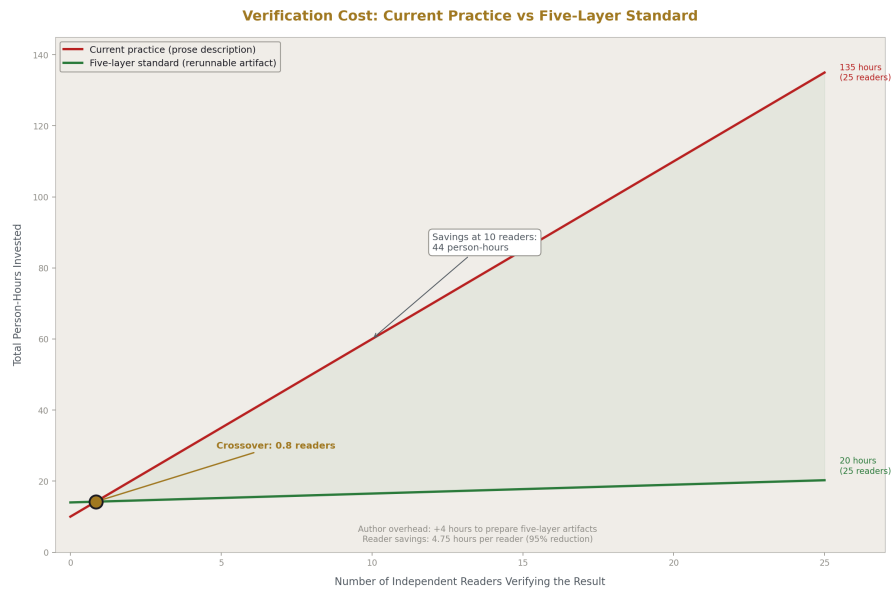


Figure 8: Fig. 5: Verification Cost — Current practice scales linearly with readers. The five-layer standard is flat after initial author investment.

8. The Standard Applied

Three categories of current scientific publication, each examined against the standard.

Theoretical computation papers. Currently: LaTeX equations, prose derivation, stated result, citations to input values. Under the standard: value store with all inputs, experiment definition with all derivations and comparisons, connection bundle stating purpose, run output with all results, generated report. The paper still exists — as the human-readable narrative explaining what was done and why it matters. But the computation exists alongside it as a rerunnable artifact. The paper describes the science. The artifact is the science. Both are published.

Experimental measurement papers. Currently: description of apparatus, description of procedure, stated result with uncertainty, data availability statement. Under the standard: value store with calibration and configuration, measurement protocol, semantic context, raw timestamped data at native sampling rate, reduction pipeline as executable code, generated report. The paper still exists as narrative. The data, the pipeline, and the results exist as rerunnable artifacts.

Reference compilations. Currently: committee-averaged recommended values published on multi-year cycles. Under the standard: every contributing instrument's time series with full metadata, the averaging methodology as executable reduction pipeline, the recommended values as pipeline output. Anyone can run the pipeline. Anyone can modify the methodology. Anyone can query the raw data. The committee's role shifts from producing the answer to maintaining the infrastructure and certifying data quality.

In every case, the standard adds a layer of verifiable artifact to an existing layer of narrative description. Nothing is taken away. The paper, the prose, the narrative explanation — all remain. What is added is the computation itself, in a form that does not require manual re-execution to verify.

9. Falsification Conditions

This paper practices what it prescribes.

Prediction 1: Reproducibility. If two independent groups implement the five-layer standard and run the same experiment definition against the same value store, they will produce identical results. If implementation differences produce different results from the same declared inputs and derivations, the standard is insufficiently specified and requires revision at the point of divergence.

Prediction 2: Error detection. If the five-layer standard is applied to a published theoretical result and the rerunnable artifact produces a different value than the published paper states, the standard has identified an error in the published result. If this never occurs across a sufficient sample, current prose-based practice is more reliable than this paper claims.

Prediction 3: Information preservation. If the standard is applied to experimental measurements and the raw data plus reduction pipeline reveals no information beyond what the published averaged result contains — no temporal structure, no contextual correlations, no instrument-specific patterns — then the averaging process does not destroy

scientifically valuable information and the additional infrastructure is not justified for experimental data.

Prediction 4: Anti-smuggling completeness. If a computation conforming to the standard nevertheless contains undeclared inputs that affect the result, the standard’s structural prevention of smuggling is incomplete and requires architectural revision.

Each test is specific. Each is observable. Each could fail. The paper commits and accepts falsification if it comes.

10. Closing

The reference implementation exists. Over 4900 value nodes. Fifty experiments. Domains spanning quantum electrodynamics, electroweak physics, cosmology, Big Bang nucleosynthesis, CKM mixing, confinement, gravity, and Hubble tension. Each experiment with pre-registered falsification conditions. Each run producing a complete audit trail. Each failure located precisely.

The BBN experiment starts from gauge integers in particle physics and derives primordial nuclear abundances in cosmology. The derivation crosses three domains. The predictions are compared against Planck satellite measurements and precision spectroscopy of high-redshift absorbers. The deuterium prediction matches to three digits. The helium prediction misses the digit target — and the report says so, in print, with the exact miss percentage, because the falsification condition was pre-registered in the experiment definition and the runner evaluated it mechanically.

That is what showing your work looks like. Not “our result is consistent with observation.” Not “we obtain reasonable agreement.” The predicted value is 0.248643. The measured value is 0.2449. The miss is 1.53%. The sigma is 0.94. The digit test fails. The sigma test passes. Every number printed. Every comparison evaluated. Every failure located. Every input traceable. Every step rerunnable.

The infrastructure to do this is commodity technology. Value stores are databases. Experiment definitions are structured data files. Runners are programs. Reports are generated text. None of this is new technology. None of this is expensive. None of this is impractical. It is a choice.

The current practice of describing computations in prose and asking readers to re-derive them manually is a convention inherited from an era of printed pages and pen-and-paper calculation. The convention persists because conventions persist — not because the constraints that created the convention still exist. The constraints are gone. The convention remains.

Showing your work means publishing the work. The inputs. The definitions. The outputs. The comparisons. The failures. All of it, in a form that anyone can run, anyone can check, and anyone can extend. In 2026, the infrastructure to meet this standard is available to every researcher with a computer.

Everything else is telling, not showing.

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Series cross-references (for deeper treatment of concepts introduced in this paper):

- Statistical breadth substituting for functional depth: [\[HOWL-CULT-1-2026\]](#)
 - What falsification structurally requires: [\[HOWL-CULT-4-2026\]](#)
 - Why located errors are the most valuable finding: [\[HOWL-CULT-5-2026\]](#)
 - Publications as immutable timestamps: [\[HOWL-CULT-6-2026\]](#)
 - The structural mechanics of institutional non-commitment: [\[HOWL-CULT-12-2026\]](#)
 - Replacing averaged snapshots with synchronized measurement: [\[HOWL-CULT-13-2026\]](#)
 - Content-based evaluation of scientific contributions: [\[HOWL-CULT-14-2026\]](#)
-

CULT-15 Supporting Appendices

Appendices to: Showing Your Work in 2026

Purpose: Reference tables consolidating the five-layer specification, the reference implementation anatomy, failure mode analysis, cost analysis, and adoption checklist. These appendices serve as the actionable engineering reference for implementing the standard.

Appendix A — The Five-Layer Specification

A.1 Layer Summary

Layer	Name	Purpose	Format	Authored By	When Created
1	Value Store	Single source of truth for all inputs	Named, typed, versioned value nodes	Researcher + community	Incrementally, reused across experiments
2	Experiment Definition	Pre-registered commitment to inputs, derivations, outputs, and falsification conditions	Machine-readable structured document	Researcher	Before computation runs
3	Connection Bundle	Semantic manifest declaring purpose, scope, and cross-domain relationships	Machine-readable structured document	Researcher	Before computation runs
4	Run Output	Complete set of results as value nodes	Same format as Layer 1	Runner (automated)	During computation
5	Report	Human-readable summary with located failures	Generated text	Runner (automated)	After computation

A.2 Value Store Node — Required Fields

Field	Type	Required	Purpose	Example
key	string	yes	Unique machine-readable identifier	cosmo yp measured v0
canonical	string	yes	Version-independent identifier	cosmo yp measured
version	integer	yes	Schema version for the node	0
node type	enum	yes	Structural classification	value
value	string, fraction, integer, boolean	yes	The datum itself	0.2449
value type	enum	yes	Representation category	approximate, exact fraction, exact integer
unit	string	yes	Physical unit or dimensionless	dimensionless, MeV, K, GeV^4
source	string	yes	Publication, year, specific table or equation	Aver, Olive, Skillman 2015. JCAP 07, 011

A.3 Value Store Node — Recommended Fields

Field	Type	Purpose	Example
topic	string	Domain classification for search	cosmo, qed, ckm, bbn, si
term	string	Physical quantity name	yp measured, boltzmann constant

Field	Type	Purpose	Example
level	integer	Hierarchy level (0=substrate, 1=theoretical, 2=measured, 3=derived)	2
digits	integer	Significant figures in value	4
uncertainty	string	Measurement uncertainty	0.0040
tags	array of strings	Searchable classification	["cosmology", "BBN", "helium", "measured"]
notes	string	Methodological context, caveats, relationships	Y p = primordial 4He mass fraction...

A.4 Experiment Definition — Required Sections

Section	Purpose	Contents
key	Unique identifier for the experiment	experiment bridge bbn v0
description	Human-readable purpose statement	What the experiment tests and why
purpose	Program-level classification	Which research program this experiment belongs to
dependencies.values	Declared inputs	Every value store key the computation requires
dependencies.derivations	Declared computations	Every derivation the experiment will execute
execution plan	Ordered derivation list	Derivations in execution order
expected outputs	Pre-declared results	Every output key the derivation should produce
comparisons	Pre-registered falsification conditions	Every test with match mode, expected value, tolerance, reference

A.5 Comparison Entry — Required Fields

Field	Type	Purpose	Example
label	string	Human-readable description of test	D/H digits of agreement
output key	string	Which result to test	result dh derived v0
match mode	enum	How to evaluate	miss pct, digits, range, bool
expected	string or number	Reference value or target	2.53e-5
reference source	string	Where expected value comes from	Cooke 2018

A.6 Comparison Match Modes

Mode	Evaluates	Pass Condition	Failure Output
miss pct	Percentage deviation from expected	Informational — reports miss	Miss percentage with digit-by-digit comparison
digits	Agreement to N significant figures	First N digits match	Specific digit position where divergence occurs
range	Value falls within [lo, hi]	$lo \leq \text{value} \leq hi$	Value and which bound was violated
bool	Boolean identity	$\text{value} == \text{expected}$	Expected vs got

A.7 Connection Bundle — Required Fields

Field	Type	Purpose	Example
key	string	Unique identifier	connection pctrm substrate identities v0
connection type	string	Category of connection	substrate, bridge, consistency
description	string	What the bundle tests and why	Full scope description
inputs	array of strings	Every value store key used	All twenty-eight input keys listed

Field	Type	Purpose	Example
tags	array of strings	Searchable classification	["PCTRM", "sub-strate", "cross domain"]
notes	string	Scope and hierarchy levels covered	Which hierarchy levels, which domains

A.8 Run Output Node — Required Fields

Field	Type	Purpose	Example
key	string	Run-specific unique identifier	experiment bridge bbn v0 run011 result omega b derived v0
canonical	string	Version-independent identifier	experiment bridge bbn v0 run011 result omega b derived
version	integer	Schema version	0
node type	enum	Always value	value
value	string or number	The computed result	0.049035637966613
value type	enum	Representation category	approximate
unit	string	Physical unit	dimensionless
source	string	Which run produced this value	experiment bridge bbn v0 run011

A.9 Report — Required Sections

Section	Contents	Generated From
Header	Experiment name, result file, timestamp, status, mode, purpose	Experiment definition + run metadata
Derivation Outputs	Every output value with its computed result	Layer 4 run output

Section	Contents	Generated From
Comparisons	Every test with predicted, measured, agreement, miss, status	Layer 2 comparisons evaluated against Layer 4 output
Summary	Derivation count (OK/error), comparison count (PASS/FAIL/INFO/SKIP), overall status	Aggregated from above

Appendix B — BBN Experiment Anatomy

Complete structural decomposition of the reference cross-domain experiment.

B.1 Input Values — Full Catalog

Key	Value	Unit	Digits	Source	Domain
geom pi v0	π	dimensionless	exact	Mathematical constant	Mathematics
geom zeta3 v0	$\zeta(3) = 1.20206\dots$	dimensionless	exact	Mathematical constant	Mathematics
integer yang mills eleven v0	11	dimensionless	exact	Gauge structure	Particle physics
integer b2 mod- ified numera- tor abs v0	13	dimensionless	exact	Gauge structure	Particle physics
cosmo omega dm planck v0	0.2607	dimensionless	4	Planck 2018	Cosmology
cosmo omega b planck v0	0.0490	dimensionless	3	Planck 2018	Cosmology

Key	Value	Unit	Digits	Source	Domain
cosmo omega de planck v0	0.6889	dimensionless	4	Planck 2018	Cosmology
cosmo omega m planck v0	0.3111	dimensionless	4	Planck 2018	Cosmology
cosmo dm to baryon planck v0	5.3204	dimensionless	5	Planck 2018	Cosmology
cosmo dm to baryon ratio prefac- tor v0	$22 \pi / 13$	dimensionless	exact	Derived	Cross- domain
cosmo t cmb v0	2.7255	K	5	Fixsen 2009	Cosmology
cosmo n gamma cmb v0	410.7	cm^{-3}	4	From T CMB	Cosmology
cosmo rho crit v0	$8.53\text{e-}30$	g/cm^3	3	From H0 Planck	Cosmology
cosmo eta planck v0	$6.104\text{e-}10$	dimensionless	4	Planck 2018	Cosmology
cosmo yp mea- sured v0	0.2449	dimensionless	4	Aver et al. 2015	Nuclear/observational
cosmo dh mea- sured v0	$2.527\text{e-}5$	dimensionless	4	Cooke et al. 2018	Nuclear/observational

Key	Value	Unit	Digits	Source	Domain
cosmo neff stan- dard v0	3.044	dimensionless	4	de Salas & Pastor 2016	Particle physics
cosmo neff planck v0	2.99	dimensionless	3	Planck 2018	Cosmology
cosmo rho lambda mea- sured v0	5.88e-30	g/cm ³	3	From Ω DE $\times \rho$ crit	Cosmology
cosmo rho lambda gev4 v0	2.85e-47	GeV ⁴	3	Converted	Cosmology
cosmo h0 planck v0	67.4	km/s/Mpc	3	Planck 2018	Cosmology
bbn yp a coeff v0	0.2485	dimensionless	4	Pitrou et al. 2018	Nuclear physics
bbn yp b coeff v0	0.0016	dimensionless	2	Pitrou et al. 2018	Nuclear physics
bbn dh a coeff v0	2.57	dimensionless	3	Pitrou et al. 2018	Nuclear physics
bbn dh b coeff v0	-0.44	dimensionless	2	Pitrou et al. 2018	Nuclear physics
astro gravi- ta- tional con- stant v0	6.674e-11	m ³ /kg/s ²	4	CODATA 2018	Gravitation

Key	Value	Unit	Digits	Source	Domain
mass proton v0	938.272...	MeV	6	PDG 2024	Particle physics
si speed of light v0	299792458	m/s	exact	SI definition	Metrology
si planck con- stant v0	6.62607015e- 34	J·s	exact	SI 2019	Metrology

B.2 Derivation Chain

Step	Derivation	Inputs From	Outputs	Domain Crossed
1	bridge omega b from integers v0	Gauge integers (11, 13), π	Ω b, Ω DE, flatness	Particle physics → Cosmology
2	bridge omega de from flatness v0	Ω b, Ω DM	Ω DE, flatness sum, residual	Cosmology (internal)
3	bridge eta from omega b v0	Ω b, T CMB, ρ crit, n γ , H_0 , G, m p, c , $\hbar$	η , η 10, n γ computed, ρ crit computed	Cosmology → Nuclear physics
4	bridge yp from eta v0	η 10, BBN Y p coefficients, Y p measured	Y p derived, miss, sigma	Nuclear physics (BBN)
5	bridge dh from eta v0	η 10, BBN D/H coefficients, D/H measured	D/H derived, miss, sigma	Nuclear physics (BBN)
6	bridge neff con- sistency v0	Ω b, Ω DM, Ω DE, N eff standard, N eff Planck	N eff check, miss	Cosmology (consistency)

Step	Derivation	Inputs From	Outputs	Domain Crossed
7	bridge vacuum energy v0	Ω DE, ρ crit, ρ Λ measured	ρ Λ derived (g/cm ³ , GeV ⁴), miss, CC ratio	Cosmology $\rightarrow$ Quantum field theory

B.3 Pre-Registered Comparisons — Full Catalog

#	Label	Output Key	Mode	Expected	Source	Result	Status
1	Ω b from integers vs Planck	result omega b de- rived v0	miss pct	0.0490	Planck 2018	0.049036, miss 0.073%	INFO
2	η derived vs Planck	result eta de- rived v0	miss pct	6.104e- 10	Planck 2018	6.090e- 10, miss 0.237%	INFO
3	η 10 derived vs Planck	result eta10 de- rived v0	miss pct	6.104	Planck 2018	6.090, miss 0.237%	INFO
4	Y p from integers vs mea- sured	result yp de- rived v0	miss pct	0.2449	Aver 2015	0.2486, miss 1.528%	INFO
5	Y p digits of agree- ment	result yp de- rived v0	digits(2)	0.245	Aver 2015	0.249, diverge at position 4	FAIL
6	D/H from integers vs mea- sured	result dh de- rived v0	miss pct	2.527e- 5	Cooke 2018	2.531e- 5, miss 0.143%	INFO

#	Label	Output Key	Mode	Expected	Source	Result	Status
7	D/H digits of agree- ment	result dh de- rived v0	digits(3)	2.53e-5	Cooke 2018	2.531e- 5, 3-digit match	PASS
8	N eff consis- tency with N eff = 3	result neff check v0	range [2.5, 3.5]	—	SM pre- diction	2.712, in range	PASS
9	N eff vs stan- dard 3.044	result neff check v0	miss pct	3.044	de Salas 2016	2.712, miss 10.9%	INFO
10	Vacuum energy (GeV ⁴) vs ob- served	result rho lambda gev4 de- rived v0	miss pct	2.85e- 47	Planck 2018	2.538e- 47, miss 10.94%	INFO
11	Vacuum energy (g/cm ³) vs ob- served	result rho lambda de- rived v0	miss pct	5.88e- 30	Planck 2018	5.889e- 30, miss 0.152%	INFO
12	Full chain: integers → η → Y p within 1 σ	result yp sigma v0	range [0, 2.0]	—	Aver 2015	0.936 σ , in range	PASS
13	Full chain: integers → η → D/H within 1 σ	result dh sigma v0	range [0, 2.0]	—	Cooke 2018	0.120 σ , in range	PASS

B.4 Derivation Output Values — Full Catalog

Key	Value	Description
result omega b derived v0	0.049035637966613	Baryon density from integers
result omega b measured v0	0.049	Planck reference
result omega b miss pct v0	0.0727	Miss percentage
result omega b used v0	0.049035637966613	Value used downstream
result omega de derived v0	0.690264362033387	Dark energy density from flatness
result omega de measured v0	0.6889	Planck reference
result omega de miss pct v0	0.198	Miss percentage
result omega dm input v0	0.2607	Dark matter density (input)
result omega m derived v0	0.309735637966613	Total matter density
result omega m measured v0	0.3111	Planck reference
result omega m miss pct v0	0.439	Miss percentage
result flatness sum v0	1.0	$\Omega_b + \Omega_{DM} + \Omega_{DE}$
result flatness residual v0	0.0	Departure from unity
result dm baryon ratio used v0	5.31654141376734	DM/baryon ratio ($22\pi/13$)
result eta derived v0	6.08953448897262e-10	Baryon-to-photon ratio
result eta measured v0	6.104e-10	Planck reference
result eta miss pct v0	0.237	Miss percentage
result eta10 derived v0	6.08953448897262	$\eta \times 10^{10}$
result eta10 measured v0	6.104	Planck reference
result eta10 used v0	6.08953448897262	Value used in BBN
result n gamma computed v0	410726847.924845	Photon number density (m^{-3})

Key	Value	Description
result rho crit computed v0	8.53145574533961e-27	Critical density (kg/m ³)
result h0 si v0	2.18405703175632e-18	H0 in SI (s ⁻¹)
result yp derived v0	0.248643255182356	Primordial helium fraction
result yp measured v0	0.2449	Aver 2015 reference
result yp measured used v0	0.2449	Value used for comparison
result yp miss pct v0	1.528	Miss percentage
result yp sigma v0	0.935813795589048	Deviation in sigma units
result yp a used v0	0.2485	BBN fitting coefficient a
result yp b used v0	0.0016	BBN fitting coefficient b
result yp at neff3 v0	0.248643255182356	Y p at N eff = 3
result dh derived v0	2.53060482485205e-5	Primordial deuterium ratio
result dh measured v0	2.527e-5	Cooke 2018 reference
result dh miss pct v0	0.143	Miss percentage
result dh sigma v0	0.120160828401573	Deviation in sigma units
result dh x1e5 derived v0	2.53060482485205	D/H × 10 ⁵
result dh x1e5 measured v0	2.527	D/H × 10 ⁵ reference
result dh a used v0	2.57	BBN fitting coefficient a
result dh b used v0	-0.44	BBN fitting coefficient b
result neff check v0	2.71205729366492	Derived N eff from Ω ratios
result neff standard v0	3.044	Standard model prediction
result neff planck v0	2.99	Planck measurement
result neff miss standard pct v0	10.905	Miss vs standard

Key	Value	Description
result neff miss planck pct v0	9.296	Miss vs Planck
result neff sigma v0	1.63495709608869	Deviation in sigma units
result rho lambda derived v0	5.88895985727292e-30	Vacuum energy (g/cm ³)
result rho lambda gev4 derived v0	2.53821856816922e-47	Vacuum energy (GeV ⁴)
result rho lambda kgm3 derived v0	5.88895985727292e-27	Vacuum energy (kg/m ³)
result rho lambda measured gcm3 v0	5.88e-30	Reference (g/cm ³)
result rho lambda measured gev4 v0	2.85e-47	Reference (GeV ⁴)
result rho lambda miss gcm3 pct v0	0.152	Miss in g/cm ³
result rho lambda miss gev4 pct v0	10.940	Miss in GeV ⁴
result omega de used v0	0.690264362033387	Ω DE used for vacuum energy
result cc problem ratio v0	3.93977103682322e+54	Cosmological constant problem ratio

Appendix C — Failure Mode Analysis

C.1 Current Practice Failure Modes

Failure Mode	Description	Frequency	Consequence	Detectable By Reader
Transcription error	Reader copies value incorrectly from cited paper	Common	Wrong downstream result	Only if reader catches own error
Notational ambiguity	Symbol means different things in different conventions	Common	Incorrect calculation	Only if reader knows both conventions
Derivation gap	Paper skips steps author considered obvious	Very common	Reader cannot verify intermediate steps	Only if reader can bridge gap independently
Unstated numerical choice	Step count, precision, convergence threshold not reported	Very common	Result depends on unreported choice	Not detectable from paper
Floating-point artifact	Accumulated rounding affects final digits	Common in long chains	Final digits implementation-dependent	Not detectable from paper
Parameter smuggling	Undeclared value enters computation	Occasional, usually unintentional	Result depends on hidden input	Not detectable from paper
Post-hoc comparison adjustment	Success criteria adjusted after seeing result	Occasional	False positive	Not detectable from paper
Selective reporting	Failures omitted from published results	Occasional	Incomplete picture	Not detectable from paper

C.2 Five-Layer Standard Prevention Mechanisms

Failure Mode	Prevention Mechanism	Layer
Transcription error	Values read from store by key, not transcribed	Layer 1
Notational ambiguity	Values have explicit units, types, and definitions	Layer 1

Failure Mode	Prevention Mechanism	Layer
Derivation gap	Derivation is executable code, every step runs	Layer 2
Unstated numerical choice	Configuration stored in value store with source and notes	Layer 1
Floating-point artifact	Exact fraction storage where possible, stated precision where not	Layer 1
Parameter smuggling	Runner loads only declared dependencies, undeclared access fails	Layers 1+2+3
Post-hoc comparison adjustment	Comparisons pre-registered in experiment definition before run	Layer 2
Selective reporting	Report generated from all comparisons, not written selectively	Layer 5

C.3 Remaining Failure Modes Under the Standard

Failure Mode	Description	Mitigation
Wrong value in store	Value entered incorrectly at data entry	Source field enables verification against original publication
Wrong derivation code	Computation itself contains a bug	Independent reimplementation against same definition
Incomplete comparison set	Relevant test not included in definition	Peer review of experiment definition, not just results
Infrastructure bug	Runner evaluates comparisons incorrectly	Independent runner implementations
Value store corruption	Data loss or modification	Versioning, checksums, immutable append-only storage

Appendix D — Cost Analysis

D.1 One-Time Costs

Component	Effort	Skills Required	Reusable Across
Runner implementation	2-4 weeks engineering	Python or equivalent, JSON processing	All experiments
Value store schema	1-2 days design	Data modeling	All experiments
Report generator	2-3 days engineering	Text formatting, arithmetic	All experiments
Documentation of standard	1 week writing	Technical writing	All adopters

D.2 Per-Experiment Costs

Activity	Effort (New Store)	Effort (Established Store)	Current Practice Equivalent
Value entry	1-2 hours for 28 inputs	10-20 minutes (most values already exist)	Looking up values in papers (same time, not reusable)
Experiment definition	1-2 hours	30-60 minutes (patterns established)	Informal planning (same time, not documented)
Connection bundle	30 minutes	15 minutes	No equivalent (purpose unstated)
Derivation code	Same as current practice	Same as current practice	Same — computation must be written regardless
Running experiment	Seconds (automated)	Seconds (automated)	Manual re-derivation by each reader (hours)
Reviewing report	10-20 minutes	10-20 minutes	Manual comparison against references (hours)

D.3 Amortization Profile

Store Size	Experiments	New Values Per Experiment	Reuse Rate	Marginal Cost Trend
0-100 nodes	1-5	20-30	0-20%	High (building store)
100-500 nodes	5-15	10-15	40-60%	Moderate (growing reuse)

Store Size	Experiments	New Values Per Experiment	Reuse Rate	Marginal Cost Trend
500-2000 nodes	15-30	5-10	70-85%	Low (most values exist)
2000-5000 nodes	30-50	2-5	90-95%	Minimal (mature store)

D.4 Cost Comparison — Single Experiment Lifecycle

Activity	Current Practice	Five-Layer Standard
Author: compute result	Hours-days	Hours-days (same)
Author: write paper	Days-weeks	Days-weeks (same, paper still written)
Author: prepare standard artifacts	Not done	2-4 hours additional
Reader 1: verify result	Hours (manual re-derivation)	Minutes (rerun experiment)
Reader 2: verify result	Hours (manual re-derivation, again)	Minutes (rerun experiment)
Reader N: verify result	Hours (manual re-derivation, again)	Minutes (rerun experiment)
Total verification cost (10 readers)	~50 hours manual labor	~1.5 hours + 4 hours author preparation

Appendix E — Domain Coverage of Reference Implementation

E.1 Experiment Inventory by Domain

Domain	Experiment Count	Example Experiments
QED / anomalous magnetic moments	8	alpha em killing spree,muon g2, laporta *
Electroweak	5	ew oneloop *,ew v2,electroweak anatomy,sin2 theta w *
CKM mixing	1	ckm cd mixing
Cosmology	4	cosmology chain, toroidal dm, hubble running *

Domain	Experiment Count	Example Experiments
Big Bang nucleosynthesis	2	bridge bbn, bbn extended
Cross-domain bridges	2	bridge bbn, bridge ew cosmo
Confinement / nuclear	1	confinement boundary
Gravity / relativity	3	soliton gravity, gr time dilation, relativity
Mathematical foundations	3	math11 beta metric, beta content a3, beta unification
Hydrogen spectroscopy	1	hydrogen 1s2s
Parameter reduction	2	parameter reduction, pctrm b round 0
Proton decay	1	proton decay
Diagnostic / scanning	5	whatif *, giga remainder test
Total	50	—

E.2 Value Store Coverage by Domain

Topic	Approximate Node Count	Examples
Cosmology (cosmo)	~400	Ω parameters, H_0 values, CMB, BBN abundances
QED (qed)	~600	α contributions by loop order, g-2 components
Electroweak (ew)	~300	Z widths, mixing angles, radiative corrections
CKM (ckm)	~100	Matrix elements, unitarity
Masses (mass)	~200	All SM particles, ratios
Couplings (coupling)	~200	α_{em} , α_s , Fermi constant
Geometry / math (geom, metric)	~300	π , β , ζ values, elliptic integrals
SI constants (si)	~50	c, $\hbar$, k B, e, N A
BBN (bbn)	~50	Fitting coefficients, measured abundances
Configuration (config)	~30	Numerical parameters
Experiment results	~3000+	Outputs from all 50 experiments
Total	~4900+	—

Topic	Approximate Node Count	Examples
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Appendix F — Adoption Checklist

F.1 Minimum Viable Implementation

Step	Action	Validates
1	Define value store schema (key, value, type, unit, source minimum)	Layer 1
2	Enter input values for one computation with full metadata	Layer 1 operational
3	Write experiment definition with dependencies, plan, outputs, comparisons	Layer 2 operational
4	Write connection bundle listing all inputs and stating purpose	Layer 3 operational
5	Implement runner that loads store, executes derivation, evaluates comparisons	Layers 4+5 operational
6	Run experiment, inspect output values and report	Full pipeline operational
7	Have a second person run same experiment against same store	Reproducibility confirmed

F.2 Quality Criteria for Each Layer

Layer	Quality Test	Pass Condition
1 — Value Store	Every input traceable to source	Source field populated, verifiable against original publication
1 — Value Store	No duplicate entries	Each physical quantity has exactly one canonical key
1 — Value Store	Exact where possible	SI-exact constants stored as exact fractions, not floats

Layer	Quality Test	Pass Condition
2 — Experiment Definition	All dependencies declared	Runner loads only declared values, no undeclared access
2 — Experiment Definition	Comparisons pre-registered	All comparisons written before run, not added after
2 — Experiment Definition	Expected outputs enumerated	Every output the derivation produces is declared in advance
3 — Connection Bundle	All inputs listed	Bundle input list matches experiment dependency list
3 — Connection Bundle	Purpose stated	Description explains what is being tested and why
4 — Run Output	All outputs stored	Every declared expected output has a corresponding output node
4 — Run Output	Source traceability	Every output node references the specific run that produced it
5 — Report	Generated not written	Report produced by runner from output, not authored separately
5 — Report	All comparisons included	No comparison from definition omitted in report
5 — Report	Failures printed	FAIL status printed with same precision as PASS status

F.3 Scaling Path

Phase	Scope	Store Size	Experiment Count	Team Size
Proof of concept	One computation, one domain	50-100 nodes	1-3	1 person
Single domain	All computations in one subfield	200-500 nodes	5-15	1-3 people
Cross-domain	Computations spanning multiple subfields	500-2000 nodes	15-30	3-5 people

Phase	Scope	Store Size	Experiment Count	Team Size
Comprehensive	All major computations in a research program	2000-5000 nodes	30-50+	5-10 people
Community standard	Multiple groups sharing store and definitions	10000+ nodes	100+	Open community

Appendix G — Pre-Registered Falsification Structure

G.1 How Pre-Registration Is Embedded

Current Practice	Five-Layer Standard
Pre-registration filed with external registry	Pre-registration is the experiment definition
Separate document from the experiment	Same document that runs the experiment
Compliance checked after the fact by humans	Compliance enforced by runner at execution time
Success criteria may be ambiguous or qualitative	Success criteria are mechanical: match modes with explicit tolerances
Partial reporting possible — omit unfavorable tests	All comparisons evaluated and printed automatically
Reinterpretation of criteria possible after seeing results	Criteria are versioned data — changes are traceable

G.2 Comparison as Falsification Commitment

Element of Falsification (Popper)	Corresponding Structure in Experiment Definition
Commitment: state specific prediction	<code>output key + expected value</code>
Specificity: prediction can disagree with observation	<code>match mode + tolerance (digits, range, percentage)</code>
Priority: prediction before test	Definition file created before run; versioned with timestamp
Acceptance: accept failure when it occurs	Runner prints FAIL mechanically; report includes all failures

G.3 BBN Experiment — Falsification Surface

Test	What Would Falsify	Located At
Ω_b miss > 1%	Integer derivation of baryon density wrong	Step 1 of chain
η miss > 1%	Cosmological bridge wrong	Step 3 of chain
Y p outside 2σ	BBN chain from integers to helium fails	Steps 1→3→4
D/H outside 2σ	BBN chain from integers to deuterium fails	Steps 1→3→5
N eff outside [2.5, 3.5]	Derived Ω ratios inconsistent with three neutrino species	Step 6
Flatness residual > 0.001	Cosmic partition doesn't close to unity	Steps 1→2
ρ_Λ (g/cm ³) miss > 1%	Vacuum energy derivation wrong	Step 7

End of Supporting Appendices.

[[HOWL-CULT-15-2026](#)] Showing Your Work in 2026. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT-15-2026>

[[HOWL-CULT-1-2026](#)] The N=1000 Fallacy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-1-2026>

[[HOWL-CULT-2-2026](#)] The Domain Boundary. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-2-2026>

[[HOWL-CULT-3-2026](#)] The Falsification Forfeiture. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-3-2026>

[[HOWL-CULT-4-2026](#)] What Falsification Actually Requires. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-4-2026>

[[HOWL-CULT-5-2026](#)] Falsification as Finding. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-5-2026>

[[HOWL-CULT-6-2026](#)] Immutable and Mortal. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-6-2026>

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